KOWSIK BOOPATHI S.

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CAREER OBJECTIVE

Electronics and Communication Engineering undergraduate with a strong academic foundation and hands-on experience building real-world automation projects. Eager to apply problem-solving skills to practical engineering challenges.

EDUCATION

B.Tech — Electronics and Communication Engineering CGPA: 7.38
Vellore Institute of Technology, Vellore • 2025 – Present

BS — Data Science and Applications (Online Degree) CGPA: 6.86
Indian Institute of Technology Madras • Foundation Level

Senior Secondary Education (12th) — CBSE 84.8%
2025

Secondary Education (10th) — CBSE 77.0%
2023 80.8% (Best of 5)

SKILLS

Programming Languages: Python, MicroPython, Java, Verilog

Hardware & Embedded Systems: ESP32 Microcontrollers, Digital Logic Design, Microcontroller Interfacing, Hardware Testbenches, Sensor Integration

Tools & Frameworks: EDA Playground, Digital Logic Simulators, IoT System Design, Smart Connect Integration Platform

PROJECTS

IoT-Based Smart Door Lock System

- Designed and built a secure, web-controlled automated door-locking system using an ESP32 microcontroller and servo motor, with logic authored in MicroPython to handle Wi-Fi connectivity, user commands, and servo actuation.
- Designed detailed circuit schematics and built hardware testbenches to evaluate system performance and reliability.
- Tested the complete setup end-to-end to ensure consistent and secure remote operation.

Dual-Mode IoT Smart Water Management System

- Engineered a dual-mode microcontroller system that switches between rainwater harvesting/diversion control and biological contamination monitoring, integrating headspace gas-analysis sensors to evaluate water quality.
- Designed the mode-switching logic and sensor integration architecture to enable autonomous, low-intervention operation.
- Programmed and tested the system to ensure accurate sensing and reliable automated control.

GSM-Based Remote Motor Starter for Agricultural Pump

- Built an SMS-controlled remote starter for a 5HP farm motor using an ESP32 microcontroller, relay module, voltage sensors, and a GSM module, enabling remote start/stop without needing to be physically present at the field.
- Implemented a programmable timer to automatically start and stop the motor for scheduled irrigation cycles.
- Integrated voltage sensors to monitor 3-phase supply voltage, allowing real-time power condition checks via SMS for safer motor operation.

EXTRACURRICULAR ACTIVITIES

Member, Robotics Club
Vellore Institute of Technology (VIT), Vellore

- Collaborate with peers on robotics and embedded systems builds, contributing to hands-on projects and technical discussions.

HOBBIES & INTERESTS

Electronics tinkering — building practical IoT/GSM-based automation solutions outside of coursework.